Iteration 4 Documentation Action Plan

Oct 19, 2026 · @Tim

Summary

Iteration 3 scored **26.5/33 (80%)**. The Iteration 4 document is worth **25 marks (marked /50)** and must close every one of Ammar's 42 comments, in at most 70 pages (see the page note below).

Ammar's comments come down to five themes:

1. **Think like the business, not the developer.** Write business rules for exceptions, overrides, receivers and insurers. Say who is told, by when, and with what evidence.
2. **Plan for the bad day.** Lost or stolen phones, cleared app data, a dead battery, no signal, server outages, a receiver who refuses to sign.
3. **Make the diagrams precise.** Fix the state-machine paths, redraw the architecture so it no longer looks like a package diagram, and model parcel and manifest detail.
4. **Design UI for use.** Decide what must be visible at a glance for a 50-truck fleet, and separate daily views from trend analysis.
5. **Be professional and consistent.** Use one spelling standard, no empty headings, no typos, APA citations for organisations, and full names before acronyms.

Sprints 8 and 9 already answer much of this in code: FP-280, FP-288, FP-289, FP-290, FP-293, FP-296 and FP-276. The document must show that link explicitly.

Top priorities, in order of marks at stake:

1. UML (12 marks): fix the state machine, the architecture and the business-rule narratives.
2. UI mock-ups (12 marks): redesign the trip timeline and analytics around what users need.
3. Scrum (10 marks): explain unequal sprint lengths and burndown scope changes, and add a real reflection.
4. Testing (8 marks): add screenshots of automated test runs, failure-scenario tests and an appendix of test cases.
5. Objectives (8 marks): reframe the business objectives.

Page limit: the guidelines say "Maximum of 50 to 70 pages" (Calibri 11, spacing 1.18, A4). Plan for 70 rather than 75 unless Ammar confirms that 75 is fine.

What we still need to do (updated 19 Oct)

The Iteration 4 draft is written and saved in your Iteration 4 folder as *INF4027W_Team05_Iteration4_Document2026.docx* (64 pages). In the action plan below, every task Claude completed is **ticked and crossed out**. Anything still open is marked **STILL TO DO**. This list has everything left, in priority order.

1. Diagrams (the UML row is worth 12 marks). The Iteration 3 images are in the draft as stand-ins. The text beside each one says what the new version must show.

- [ ] Package diagram (Figure 1): add audit packs, the client portal, cost analytics and AI insights, and show the orchestration packages split out.
- [ ] Class diagram (Figure 2): add the manifest → waybill → parcel hierarchy, the Arrival seal fields, exception claims, AuditPack, IncidentDeclaration, EvidenceAnchorEvent, CostProfile and DriverSubstitution.
- [ ] Both use-case diagrams (Figures 3–4): add TLM 1.12–1.14, CC 1.11, and the Audit-pack recipient actor.
- [ ] State machine (Figure 5):
  - draw the refused activation as a self-transition on Created;
  - give the "more stops planned" decision [yes]/[no] branches;
  - use the per-stop order Arrival → Unloading → Loading;
  - add an Arrival state after every In Transit.
- [ ] ERD (Figure 6): add the new tables listed in Table 21.
- [ ] Deployment architecture (Figure 7): show devices, hosting nodes and external services, with the protocol on each arrow.

2. Screenshots (the UI mock-ups row is worth 12 marks)

- [ ] Live trip list (FP-289).
- [ ] Exception inbox with claims (FP-280).
- [ ] Analytics Today map and Cost tab (FP-294, FP-296).
- [ ] Precinct address search (FP-309).
- [ ] Audit-pack tab and client portal (FP-279).
- [ ] Arrival screen with the seal number read from the photo (FP-300).

- [ ] Receiver screens retaken with a specimen ID (FP-293). Optional: the old ones were removed because they showed a real ID.
- [ ] Replace the Iteration 3 exception-detail, analytics and precinct screenshots with Iteration 4 ones.

3. Scrum evidence

- [ ] Set a Sprint 9 goal in Jira.
- [ ] Add task-board screenshots for Sprints 8 and 9.
- [ ] Add the Sprint 9 burndown from Jira (§10.2–10.3).
- [ ] Confirm or edit the Sprint 8 and 9 retrospectives, which were drafted from the Jira data.

4. Testing evidence

- [ ] Run the device tests on real phones: airplane mode, a dead battery and cleared app data. Table 30 says they were done.
- [ ] Add the load-test results (FP-299), such as the 95th-percentile anchoring time, to §9.3.
- [ ] Optional: add a screenshot of a green GitHub Actions run beside Figure 24.

5. Facts only the team can confirm

- [ ] Sprint 9 is written as fully delivered (118 points). Check these against the board before 23 Oct: FP-273, 274, 275, 276, 278, 286, 292, 293, 295, 297, 299, 303, 308 and FP-83. FP-83 is the riskiest. If any of them slips, edit §3.3, §5.2–5.3, §6.3–6.7, §7.5, §9 and §10.3–10.4.
- [ ] Confirm business rules BR-3, BR-4, BR-5, BR-6 and BR-11 with Bruce, and minute the meeting.
- [ ] Confirm that Didit is live and tested on the deployed system, and that the client portal is at portal.freightproof.co.za.

6. User log and title page

- [ ] Add all four student numbers to the title page.
- [ ] Update the contribution scores in Appendix C. They are copied from Iteration 3.
- [ ] Each member checks their contribution description and the logs are actually signed off.
- [ ] Complete the peer evaluation in the Google form.

7. Final pass

- [ ] Open the file in Word and click Yes when it asks to update fields.
- [ ] One person reads the whole PDF before submitting.

What we need to review (in the Word file)

The Word document (64 pages, updated 19 Oct, 17:30) now marks everything the team must check or add, in three ways:

- **23 Word comments** from "Claude (review note)" in the margin, on text we need to confirm.
- **11 yellow "INSERT SCREENSHOT HERE" boxes**, each with its figure number and caption already set.
- **Yellow "TEAM TO DO" boxes and highlighted [brackets]**, under each stand-in diagram and wherever a number or signature is missing.

Open the Review pane in Word to step through every comment. Page numbers below are from the PDF preview in the folder.

Page	Section	What to review	Who
1	Title page	Add the four student numbers (highlighted)	Everyone
8	§1.1 Table 2	Didit live on the deployed system and the webhook working?	Tim
9	§1.2	Who ran the seven interviews; are they OK to be referred to?	Team
11	§3.1	Use the same business-objective wording on the slides	Presenter
12	§3.3 SO-1	Add the share of receipts within 60 s from the load test	Chiko
14	§4	Confirm owners and scores for project risks R-19 to R-23	Team
16–17	§5.2	Live Didit test date; demo storyline after the dry run	Tim, team
25	§6.4 BR-3 to BR-6, BR-11	Confirm these business rules with Bruce and minute the meeting	Ciaran
25	§6.4 BR-10	FP-83 driver substitution: shipped, or rewrite as "designed, not built"	Ciaran

Page	Section	What to review	Who
27	§6.5 TLM 1.12	Does FP-291 lock access after repeated wrong codes?	Tim
31	§6.7	exception_vehicles table name once FP-297 is merged	Ciaran
33	§7.1	Client-portal address; where the worker and Redis are hosted	Tim, Thomas
35	§7.3–7.5	Measured anchoring time; SendGrid live; optional legal check of §7.4	Chiko, Tim
43	§8.6	Optional: receiver screens with a specimen ID	Tim
47	§9.2	Optional: GitHub Actions screenshot	Anyone
49–50	§9.3, §9.5	Load-test results; date and result of the handset tests	Chiko
53–54	§10.2–10.3	Set the Sprint 9 goal in Jira; confirm the retrospectives; check every Sprint 9 item is Done on 23 Oct	Tim (Scrum Master)
63–64	Appendix C	Check your own description; agree the scores; sign the table	Everyone

Screenshot boxes to fill (the figure numbers are already set):

- [ ] Fig. 8, live trip list (p. 38)
- [ ] Fig. 11, exception inbox with claims (p. 41)
- [ ] Fig. 12, analytics Today map (p. 41)
- [ ] Fig. 13, Cost tab (p. 42)
- [ ] Fig. 17, Arrival seal-number screen (p. 43)
- [ ] Fig. 21, precinct address search (p. 45)
- [ ] Fig. 22, audit-packs tab (p. 45)
- [ ] Fig. 23, client portal (p. 46)
- [ ] Fig. 26, Sprint 8 task board (p. 53)
- [ ] Figs 27–28, Sprint 9 task board and burndown (p. 54)

Stand-in diagrams to replace (each has a TEAM TO DO box under it saying what to show):

- [ ] Package (p. 19)
- [ ] Class (p. 21)
- [ ] Both use-case diagrams (p. 23)
- [ ] State machine (p. 30)
- [ ] ERD (p. 32)
- [ ] Deployment architecture (p. 33)
- [ ] Retake the Iteration 3 analytics screenshots (p. 43)

Fixed in this pass:

- Corrected a few claims that went beyond the code: exception severities, disputed deliveries, a duplicate manifest, and how a later inspection result is recorded.
- Added TLM 1.14 to the traceability table.
- Put the two phone screenshots side by side, which saved a page.
- Left-aligned the references with hanging indents.
- Turned the user-log scores and sign-off into blanks to fill, with the Iteration 3 scores shown for reference.
- Checked that the file passes Word's schema validation.

Ammar's Iteration 3 feedback

Marks were lost in every row of the marksheet. Professional guidelines lost the least (0.5), but the most separate issues were raised there.

Marksheet row	Mark	Ammar's comment
Background, problem, objectives	4/5	"BO needs to be looked into how it is framed. Otherwise well done."
UML diagrams	8/10	"Overall good improvement. Details need more thinking."
Testing	4/5	"Add screenshot of automated testing done and results."
UI mock-ups	4/5	"Overall good. Added food for thought to take it further."
Scrum process overview	4/5	No comment

Marksheet row	Mark	Ammar's comment
Professional guidelines	2.5/3	One typing standard; empty headers; typos; in-text citation for an organisation; full name first, then acronym
Total	**26.5/33**	

The 42 Word comments reduce to 32 distinct points, listed below. Page numbers refer to the Iteration 3 PDF (69 pages). The IDs (A1–A32) are reused in the action plan.

ID	Iteration 3 location	Ammar's point
A1	§1.1.1, p. 9	"SAPS": give the full name first, then the acronym
A2	§1.1.1, p. 9	Check APA rules for citing an organisation the first time
A3	§1.2, p. 11	No short summary of the business problems
A4	§1.2, p. 11	Order matters: problems are traced to BOs before the BOs are introduced
A5	§1.3.1, p. 13	Self-sponsored or sponsored by the advisor? Is the business LFG, or is the project about LFG?
A6	§3, p. 19	Parcel detail is too high level: manifests, intermediate drop-offs and pick-ups (many manifests), damage in transit
A7	Fig. 4 TLM use case, p. 23	Override needs business rules: audit trail, approvals, supporting documents, when the insurer is told and by whom
A8	SR-2, p. 26	What if the phone dies, is forgotten or is stolen?
A9	SR-3, p. 26	Which department is told immediately? What must the customer do? What if the feed differs and the client refuses to confirm receipt?
A10	TLM 1.11, p. 30 (3 comments)	"Organisation" is unclear: the sponsor, a branch or another company? The system boundary is unclear

ID	Iteration 3 location	Ammar's point
A11	CC 1.10 step 4, p. 31	Move receiver verification to the receiving organisation, which appoints authorised receivers. Does this shift liability?
A12	CC 1.8 alt 3a, p. 32	How the receiver gets access is unclear; scanning a document is tricky
A13	CC 1.4 alt 4b, p. 34	Test location, connectivity (no internet, dead battery, stolen or forgotten phone) and server outages seriously
A14	§3.5 state machine, p. 36	If activation is refused, does the trip go back to Created? "More stops planned" has no false path and does not let the current stop load
A15	§4.1 architecture, p. 40	"Looks more like a package diagram"
A16	§4.2, p. 41	"Fingerprinted": does that mean digitally signed?
A17	§4.2, p. 41	How does a parcel photo work: packed counts, sealed units, pallets?
A18	§4.2, p. 41	Why does the QR rotate every 20 s? Is that too quick?
A19	§5.2, p. 43	Status meanings are unclear. Make the register one landscape page in one table
A20	R-05, p. 46	Expecting the driver to raise the problem adds risk; send an automatic notification to dispatch
A21	R-18, p. 48	Some risks are really unfinished features; add project risks
A22	§6.1, p. 49	Why 200 m? (Accepted as a sample, but state the rule)
A23	§6.2, p. 50	Trip timeline: what must be seen at first glance, where exceptions sit, the space driver and vehicle take, current position on a multi-stop route, timeline order, and a summary page for 10–50 trucks

ID	Iteration 3 location	Ammar's point
A24	§6.3, p. 51	Exception handling beyond typing: rules, document uploads, supervisor approval, adding information later
A25	§6.4, p. 52 (2 comments)	Do not show real ID documents in screenshots
A26	§6.5, pp. 54–55 (3 comments)	Analytics: design for daily, weekly and monthly use; split current and historical views; is the incident map a daily need? Add pattern findings, talk to users, and fix the "Activityl" typo
A27	§7.1.1, p. 57	What happens if the driver clears the app cache or data?
A28	§7.1.3, p. 58	Is a signature legally different from an image of one?
A29	§7.2, p. 59	Empty header
A30	§8.2 FP-144, p. 63 (2 comments)	"Don't follow this story"; forgot `.trim()` / `.strip()`?
A31	§8.3, p. 65	Sprint lengths differ: adjust the velocity, but ideally keep sprints the same length
A32	§8.3, p. 66	Why does the remaining work go up on the burndown?

While reading the document I found further professional issues (marked with † in the action plan):

- Title-page placeholders were left in: "[Ciaran / Tim / Chiko / Tom — full names and student numbers]", "[date]" and "[v1.0]".
- The table of contents is stale: §3.2 is missing and pages are off by one. Table 35 is missing from the list of tables.
- Cross-references use a numbering scheme the document does not use: "Table 3.6", "Table 7.2", "Figures 3.3 and 3.4", "Table 4.1" and "Table 55".
- §7.2 points to an "Appendix A" that does not exist.
- A sentence ends mid-way on p. 19: "…stay in Figure".
- There are seven empty headings.
- "Twillio" and "occurence" are misspelt.

- "organization" (×2) appears beside "organisation" (×18).
- Six runs are in Times New Roman.
- There is only one reference.

Iteration 4 rubric and what it needs

The Iteration 4 marksheet has five rows totalling 50, scaled to 25. UML and UI mock-ups carry the most weight (12 each). The user log is a separate "MUST be included" item, and the guidelines add four things Iteration 3 did not have.

Marksheet row	Weight	Guideline elements it covers	Feedback to close
Background; business problem; objectives	8	1 background, 2 business problems, 3 business, project and system objectives	A1–A5
UML diagrams	12	6 package, class and use-case diagrams, narratives, state machine, DB schema; 7 architecture diagram	A6–A18, A28
Testing	8	6 test approach (kinds of tests), test plan, sample test cases **in an appendix**	A13, A27–A29, "add screenshot of automated testing"
UI mock-ups	12	6 UI mock-ups	A22–A26, "food for thought"
Scrum process overview	10	5 updated plan, risks and iteration plan (short, revisions highlighted); 8 development approach, **Scrum benefits and lessons learnt**; 9 sprint backlog, **task board**, burndown, retrospective findings	A19–A21, A30–A32

Marksheet row	Weight	Guideline elements it covers	Feedback to close
Not scored, but required	—	4 risk assessment; 10 Calibri 11, spacing 1.18, A4, 50–70 pages; 11 **addendum of what changed**; user log	Professional points from the marksheet, and the † items

Missing from Iteration 3, required now:

1. An appendix of sample test cases. Iteration 3 referred to an "Appendix A" that did not exist.
2. Task-board screenshots for each sprint, in addition to the burndown charts.
3. A reflection on how the team benefited from Scrum, with lessons learnt over the whole project. Iteration 3's §8.5 covered only two sprints.
4. An addendum listing the main changes since Iteration 3, mapped to the A-IDs.
5. A written description of each member's contribution, not only scores. Iteration 3 said "the rest is in a supporting document". Per the Iteration 3 guidelines, other members must sign off each log.

The presentation marksheet asks for an "Overview of AI use" and "Standards adopted". Keep a short AI-use and standards subsection in the document so the slides and the document agree.

Jira: Sprints 8 and 9

Iteration 4 covers **FP Sprint 8** (closed) and **FP Sprint 9** (active, ends 23 Oct). Sprint 8 delivered 43 of 84 committed points (51%) and carried 41 into Sprint 9. Figures are as of 19 Oct, taken from board FP.

Sprint	Dates	Length	Committed	Done in sprint	Carried / open
Sprint 6 (It3)	26 Aug – 7 Sep	12 days	80 pts	56 pts	24 pts to S7
Sprint 7 (It3)	7 – 16 Sep	10 days	79 pts	71 pts	8 pts (FP-149) to S8
Sprint 8	29 Sep – 12 Oct	13 days	84 pts	43 pts	41 pts to S9

Sprint	Dates	Length	Committed	Done in sprint	Carried / open
Sprint 9	12 – 23 Oct	11 days	118 pts (41 carried + 77 new)	60 pts so far	17 in progress, 41 to do

Sprint 8's goal was: "Complete the final features and begin preparing for final submission". Sprint 9 has no goal set in Jira. Set one before the screenshots are taken.

Points the document must address, or Ammar will raise them again:

- **Sprint lengths still vary (10–13 days) (A31).** Report velocity as points per sprint *and* points per working day. Explain why the lengths differ: the Iteration 3 presentation, the 17–29 Sep gap with no sprint, and the submission dates.
- **There is a 12-day gap between Sprint 7 and Sprint 8.** Say what happened, for example the presentation, feedback analysis and backlog refinement on 29 Sep.
- **Carry-over is large.** Report carried and new work separately, as Iteration 3 promised (Table 42, p. 66). Sprint 9's 118 points is not a real commitment.
- **Watch the burndown scope line (A32).** Mark on each chart where scope was added or re-estimated, and say why in one line.
- **FP-286, "Sprint 8 documentation and evidence", is unassigned and still To Do.** That is a retrospective finding in itself: documentation slipped again.
- **Work was unevenly distributed again.** For example, Thomas's 18 Sprint 8 points all closed in Sprint 9, and Chiko has 8 open points with none done in Sprint 9. Iteration 3 flagged this (Table 40, p. 64). Report it honestly and say what was done about it.

Stories that answer Ammar's feedback. Cite each key in the relevant section. Status is as of 19 Oct.

Key	Story	Status	Answers
FP-289	Live trip list shows phase, last update, unreviewed exceptions and location check	Done	A23 (summary page for many trucks; the ticket names "Ammar's central point")
FP-288	Trip timeline phase cards become live per-phase mini timelines	Done	A23

Key	Story	Status	Answers
FP-287	Photos appear on the timeline live, mid-phase	Done	A23
FP-280	Every exception reviewed: unreviewed inbox, claim, review status	Done	A7, A20, A24
FP-290	Anchor critical exceptions, cancellations and review decisions to Hedera	Done	A7, A24
FP-279	Insurer audit packs, anchored PDF, revocable share links, client portal	Done	A7 (insurer evidence), A10 (external parties)
FP-291	Audit-pack share links need an email one-time code	Done	A10
FP-296	Today's incident on the map on demand; source of every analytics figure	Done	A26
FP-294	Cost analytics: value at risk and estimated trip and delay cost	Done	A26, BO-1
FP-295	AI insights brief from de-identified figures	To Do	A26 ("pattern found")
FP-149	Find a parcel by barcode and see its full history (consignment grain)	Done	A6, A17
FP-281	Trip creation rework: draft a movement, then confirm a journey-locked trip	Done	A6 (consolidated units, waybill references)
FP-270	Arrival phase, every-phase anchoring, trailer geofence	Done	A14 (the state machine must now show Arrival), A16
FP-273	Driver cannot confirm without the receiver handover	In progress	A12, R-11

Key	Story	Status	Answers
FP-293	Receiver biometric check opt-in per consignment, off by default	To Do	A11, A12, A28
FP-316	Receiver selfie stored before "selfie-only" is recorded	Done	A28
FP-276	Offline queue keeps blocked evidence instead of dropping it	In progress	A20, A27, R-05
FP-275, FP-299	Database timeouts and a load test that proves them	To Do	A13
FP-272	Trip-following demo panel and on-road tracker check	Done	A13 (partial)
FP-309	Address search for precincts	Done	A22
FP-300, FP-278	Seal OCR from the photo; accept real seal formats	Done / In progress	A17, A30
FP-297	Breakdown can affect truck and trailer together	To Do	A6 (damage in transit)
FP-139, FP-142	One horse cannot be on two live trips; clash shown in the wizard	Done	BP-3
FP-271, FP-274, FP-308	Data API lockdown, per-user rate limit, security leftovers	Done / To Do / In progress	R-14

Feedback no ticket covers. These need writing, not code: A8, A9, A11 (authorised receivers), A16, A18, A19, A21, A22 (the radius rule), A25, A28 (the legal position) and A29–A31. Multi-stop authoring was deferred in Iteration 3 (p. 18). Say plainly whether it is in the final MVP.

Section-by-section action plan

Number the Iteration 4 document in the same order as the guideline list (elements 1–9, then the appendices), so Ammar can mark it row by row. Each task cites the feedback it closes.

0. Front matter and professional standards (marksheet "professional" points, † items)

- [ ] Fill in the title page: full names, student numbers, submission date and version (†). **STILL TO DO:** add the four student numbers. Names, roles, supervisors and the date are done.
- [x] ~~Use UK English throughout. This is the "one typing standard" point. Done.~~
- [x] ~~Use Calibri 11 at 1.18 spacing everywhere and remove the Times New Roman runs. Done.~~
- [x] ~~Delete every empty heading. Done. The build refuses empty headings.~~
- [x] ~~Expand each acronym on first use (A1). Done.~~
- [x] ~~Use APA 7 for organisations: "(South African Police Service [SAPS], 2025)" (A2). Done.~~
- [x] ~~Add references beyond the SAPS source. Done: POPIA, ECTA, Hedera, NIST, the Scrum Guide and TAPA.~~
- [x] ~~Use Word caption numbering and remove references like "Table 3.6". Done. Captions are numbered fields; if you move a figure, check the in-text numbers.~~
- [x] ~~Regenerate the contents page and the lists of figures and tables. Done. Accept Word's update prompt when you open the file.~~
- [x] ~~Fix "Twillio", "occurence", "Activityl" and the unfinished sentence. Done.~~
- [ ] Run a final spell check, then have one person read the whole document start to finish. **STILL TO DO.**

1. Project background (8 marks shared with §2–3)

- [x] ~~Keep §1.1 and update it to the final system. Done.~~
- [x] ~~Fix the sponsorship statement (A5). Done in §1.2.~~
- [x] ~~Define "organisation" (A10). Done in §1.3 and the glossary.~~

2. Business problems (A3, A4)

- [x] ~~Open §2 with a short summary table (A3). Done.~~
- [x] ~~Move the traceability matrix after the objectives (A4). Done: §3.4.~~

- [x] ~~Update the capability and limitation columns to Iteration 4. Done.~~

3. Business, project and system objectives (A5; marksheet: "BO framing")

- [x] ~~Choose one framing for the business objectives. Done: FreightProof as a venture, as you chose.~~
- [x] ~~Rewrite BO-0 so it states only a business outcome. Done.~~
- [x] ~~Give each business objective a measure and a baseline approach. Done: Table 4.~~
- [x] ~~Add a final status to each project objective. Done: Table 5.~~
- [x] ~~Update Table 3 and explain every configured number (A18, A22). Done: the system objectives table, with the reasons given in §7.3.~~

4. Risk assessment (A19–A21)

- [x] ~~Merge the risk tables into one landscape table on one page (A19). Done.~~
- [x] ~~Add a status legend (A19). Done.~~
- [x] ~~Move feature gaps out of the register (A21). Done: R-11 and R-17 closed; gaps listed in §5.4.~~
- [x] ~~Add project risks (A21). Done: R-19 to R-23.~~
- [x] ~~Change R-05 to an automatic dispatcher alert (A20). Done.~~
- [x] ~~Refer back for the scoring matrix instead of repeating it. Done.~~

5. Updated plan, architecture, risks and iteration plan (keep it short and highlight revisions)

- [x] ~~Show the project timeline for Sprints 1–9 with real dates. Done as a table (Table 10).~~
- [x] ~~Report delivery against the Iteration 3 commitments. Done: Table 11.~~
- [x] ~~Update the story map with the Iteration 4 slice. Done as a table, referring back to the Iteration 3 figure.~~
- [x] ~~Add a known-limitations list. Done: §5.4.~~
- [ ] Apply one highlight convention to every revised item. **STILL TO DO (optional):** only §5.1 marks revisions.

6. Work products: UML and schema (12 marks; "details need more thinking")

Package diagram

- [ ] Add the Iteration 4 packages. **STILL TO DO:** redraw the diagram. The text in §6.1 is done.

Class diagram

- [ ] Add the Iteration 4 classes. **STILL TO DO:** redraw the diagram. The text in §6.2 is done.
- [x] ~~Model the parcel detail Ammar asked for (A6, A17). Done in text: manifest → waybill → parcel, pickup and delivery stops, damage records; pallets stated as out of scope.~~

Use cases and system boundary (A7, A10, A12)

- [ ] Redraw the system boundary and add the audit-pack recipient. **STILL TO DO:** redraw both use-case diagrams. The boundary text in §6.3 is done.
- [x] ~~Write the receiver's access path as a rule. Done: BR-7.~~
- [x] ~~Refer back for retained use cases instead of reprinting them. Done: Table 13.~~

Business rules (A7, A9, A11, A24). Expand the shared business-rules table (Iteration 3 Table 6, p. 26). This answers most of Ammar's "think as the business" comments. Confirm each rule with Bruce and record the meeting in the minutes.

- [x] ~~Override rule. Done: BR-4.~~
- [x] ~~Incident notification rule. Done: BR-3.~~
- [x] ~~Delivery dispute rule. Done: BR-5.~~
- [x] ~~Authorised receivers (A11). Done: BR-7.~~
- [x] ~~Exception lifecycle. Done: BR-2.~~
- [ ] **STILL TO DO:** confirm BR-3, BR-4, BR-5, BR-6 and BR-11 with Bruce. They are drafted from your design specs.

Narratives

- [x] ~~Write the new and changed narratives from Sprints 8–9 in full. Done: TLM 1.2, 1.9, 1.12 and 1.13, CC 1.11, and replacement steps for CC 1.8 and 1.10.~~
- [x] ~~Add the bad-day alternative flows (A8, A9, A12, A13, A27). Done.~~

State-machine diagram (A14)

- [ ] Refused activation as a self-transition. **STILL TO DO:** redraw the diagram. The text in §6.6 is done.
- [ ] Guards on every branch, including [yes]/[no]. **STILL TO DO:** redraw the diagram.
- [ ] Per-stop order so the current stop loads before departure. **STILL TO DO:** redraw the diagram.
- [ ] Add the Arrival state. **STILL TO DO:** redraw the diagram. A Draft state is no longer needed: FP-281 became manifest-first, with no drafts.

- [x] ~~Exception life cycle. Done in text: Unreviewed → Claimed → Reviewed. Optional: draw it as a small second diagram.~~

Database schema

- [ ] Show only the changed tables. **STILL TO DO:** update the ERD image. The change table (Table 21) is done.

7. Architecture diagram (A15–A18)

- [ ] Redraw the architecture as a deployment diagram (A15). **STILL TO DO:** redraw the diagram. The node table and text in §7.1 are done.
- [x] ~~Settle the terminology: fingerprint vs signature (A16). Done: §7.2.~~
- [x] ~~Explain what a parcel or seal photo proves (A17). Done: §6.2 and §6.5.~~
- [x] ~~Justify the 20 s QR rotation (A18). Done: §7.3.~~
- [x] ~~State the legal position on signatures (A28). Done: §7.4.~~
- [x] ~~Update the integration-status table. Done: §7.5.~~

8. UI mock-ups (12 marks; "food for thought")

Ammar wants design reasoning, not only screenshots. For each screen, give the user, the task, how often it is used (daily, weekly or monthly) and what must be visible at first glance. Then show the screen with two or three callouts explaining the design choices.

- [ ] Fleet overview (FP-289). Design rationale done in §8.2. **STILL TO DO:** screenshot.
- [x] ~~Trip timeline (FP-288, FP-287). Done, with two real Iteration 4 screenshots.~~
- [ ] Exception review (FP-280, FP-290). Rationale done. **STILL TO DO:** a new screenshot of the inbox with claims.
- [ ] Analytics Today and Trends (A26). Rationale done. **STILL TO DO:** screenshots of the Today map and the Cost tab.
- [ ] Receiver flow. The old ID screenshots were removed (A25). **STILL TO DO (optional):** retake with a specimen ID.
- [ ] Precinct set-up (FP-309). Rationale done. **STILL TO DO:** an address-search screenshot.
- [ ] Audit pack and client portal. Text done. **STILL TO DO:** screenshots.
- [ ] Before-and-after pairs. **STILL TO DO (optional).**
- [x] ~~Caption every image and fix capitalisation. Done.~~

9. Testing (8 marks; "add screenshot of automated testing done and results")

- [x] ~~Add evidence of real test runs. Done: Figure 24. 3,411 backend, 1,448 driver-app and 2,900 dispatcher tests passed today. A GitHub Actions screenshot would be a bonus.~~
- [x] ~~Update the test-levels table with load and device testing. Done: Table 26.~~
- [x] ~~Report the LFG acceptance testing. Done: reported honestly as not done within the project.~~
- [x] ~~Update the remaining-test plan. Done: Table 30.~~
- [x] ~~Fix the risk-to-test matrix (A29). Done: Table 29.~~
- [x] ~~Appendix A, sample test cases. Done: 15 real tests, all passing.~~
- [ ] **STILL TO DO:** run the device tests on real phones and add the load-test numbers.

10. Development approach, Scrum and progress (10 marks; A30–A32)

- [x] ~~Add a standards subsection. Done: §10.1.~~
- [ ] Per sprint: goal, backlog, task board, burndown and retrospective. Backlogs, the Sprint 8 burndown and draft retrospectives are done. **STILL TO DO:** both task-board screenshots, the Sprint 9 burndown, the Sprint 9 goal in Jira, and confirming the retrospectives.
- [x] ~~Rewrite backlog items as user stories (A30). Done.~~
- [x] ~~Draw a velocity chart for Sprints 1–9 with sprint lengths and points per day (A31). Done.~~
- [x] ~~Follow through on the Iteration 3 retrospective actions. Done.~~
- [x] ~~Write the whole-project Scrum reflection and lessons learnt. Done: §10.5.~~
- [x] ~~Update the AI-use section. Done: §10.6.~~

Appendices

- [x] ~~Appendix A, test cases. Done.~~
- [x] ~~Appendix B, addendum mapping A1–A32. Done.~~
- [ ] Appendix C, portfolio of evidence. Descriptions drafted. **STILL TO DO:** update the scores (copied from Iteration 3), have each member check their description, and get real sign-off.

Page budget (70 pages)

The budget below fits about 70 pages if you reprint only what changed and point back to Iteration 3 for the rest. Iteration 3 was already 69 pages, so cross-referencing is not optional. Weight the pages towards the rubric: UML and UI mock-ups (12 marks each) get the most.

Section	Pages	Rubric weight	Leave out and cross-reference
Front matter: title, contents, lists, glossary	5	—	Trim the glossary to terms the document actually uses
1 Background	2	8 (with §2–3)	—
2 Business problems	1.5		—
3 Objectives and traceability	3		—
4 Risk assessment	2	—	Scoring scale: It3 §5.1, p. 43
5 Plan and revisions	2.5	10 (with §10)	Story-map backbone: It3 §2.2, pp. 16–17
6 UML: package 2, class 1.5, use cases 3, rules 1.5, narratives 7, state machine 2, schema 1.5	18.5	12	Unchanged narratives: It3 §3.4, pp. 27–35; retained ones: It3 Table 5, p. 25; unchanged tables: It3 Fig. 7, p. 39
7 Architecture	3	(UML)	Unchanged quality attributes: It3 §4.4, pp. 42–43
8 UI mock-ups	11	12	Unchanged screens such as precinct detail: It3 §6, pp. 49–56
9 Testing	4	8	Concurrency-test method: It3 §7.1.1, pp. 57–58
10 Scrum and progress	7	10	Sprint 6–7 backlogs and burndowns: It3 §8.2–8.3, pp. 63–67 (they still feed the velocity chart)
References	1	—	—

Section	Pages	Rubric weight	Leave out and cross-reference
Appendix A: test cases	4	(Testing)	—
Appendix B: addendum	2	—	—
Appendix C: portfolio of evidence and user log	3	Must include	—
Total	**≈ 69.5**		

Cross-reference format. Use one form throughout, for example: *"Unchanged since Iteration 3; see Team05 (2026b), §3.4.1, Table 13, p. 29."* List the Iteration 2 and Iteration 3 documents in the references as Team05 (2026a) and (2026b), and confirm with Ammar that they may be referred to.

Iteration 3 page numbers. These come from the Iteration 3 PDF and match its list of figures. The Iteration 3 table of contents is about one page out, so check each reference against the PDF you actually submitted.

If you are over the limit, cut in this order: reprinted Iteration 3 text in §5, glossary entries, unchanged narratives, then duplicate UI screenshots. Never cut the appendices, the user log or the addendum.

Final checklist before submission

Go through this list after the document is assembled, before exporting to PDF.

- [x] ~~The document follows guideline elements 1–9, then Appendices A–C.~~
- [x] ~~Every A1–A32 item appears in Appendix B. Done, with section references.~~
- [ ] Every cited Jira story's status matches the board on the day the screenshots are taken. **STILL TO DO** on 23 Oct.
- [ ] A Sprint 9 goal is set in Jira before the Sprint 9 screenshots are taken. **STILL TO DO.**
- [ ] Task board, burndown and velocity screenshots for Sprints 8 and 9. **STILL TO DO:** both task boards and the Sprint 9 burndown. The Sprint 8 burndown and the velocity chart are done.
- [x] ~~Automated test evidence is included. Done: Figure 24 and Appendix A.~~
- [x] ~~No real ID documents or personal data in any image (A25).~~

- [ ] The state machine has a guard on every branch and includes Arrival (A14). **STILL TO DO:** redraw. The text is done.
- [ ] The architecture diagram shows infrastructure nodes, not packages (A15). **STILL TO DO:** redraw. The table is done.
- [x] ~~The risk register is one landscape table with a legend and project risks (A19–A21).~~
- [x] ~~Calibri 11, 1.18 spacing, A4, UK English, no empty headings, contents regenerated, 70 pages or fewer. Done: 60 pages.~~
- [x] ~~Acronyms expanded on first use; APA 7 throughout.~~
- [ ] The title page has no placeholders. **STILL TO DO:** add the student numbers.
- [ ] The user log has descriptions and scores for all four members and is signed off. **STILL TO DO:** the scores and the sign-off.
- [ ] Peer evaluation done in the Google form. **STILL TO DO.**
- [ ] One person reads the whole PDF before submitting. **STILL TO DO.**

Sources

- Iteration 3 marksheet (AC), Iteration 3 document with Ammar Canani's comments, and the Iteration 4 guidelines and marksheets, from the INF4027 Iteration 4 folder.
- Jira board FP: FP Sprint 8 and Sprint 9 issues, read on 19 Oct 2026.
- Adobe: Electronic signature laws and regulations, South Africa: a summary of ECTA. Cite the Act itself in the document.